Bhrij Patel

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RESEARCH INTERESTS

- Test-Time Alignment and Personalization for AI Systems
- Multi-step Tasks with Tool-based Agents
- LLM-based Evaluators and Reward Modeling
- Memory and Experience Distillation for Continual Learning
- Sample Efficient Optimization for Reinforcement Learning with Sparse Rewards

EDUCATION

University of Maryland, College Park - Advisor: Dinesh Manocha
Ph.D. in Computer Science
Thesis-in-Progress: ***Reliable Reinforcement Learning from Sparse Rewards***
Master's in Computer Science — May 2024
Duke University - Advisor: Cynthia Rudin
Bachelor of Science in Computer Science & Mathematics, Minor in Creative Writing — May 2022

WORK EXPERIENCE

Adobe Research — June 2026 - Aug. 2026
Research Intern, Bengaluru, India, Mentors: Soumyabrata Pal, Ananya Sai, Ramasuri Narayanam

- Researching synthetic data generation for multimodal, tool-based agents

AG2 (formerly AutoGen) — Nov. 2025 - Feb. 2026
Research Intern, Remote, Mentor: Huazheng Wang

- Researched continual learning and feedback generation for tool-based, multi-turn dialogue tasks
- Investigated the pitfalls of continual learning methods when past experiences are noisy and misleading

Qualcomm Research, Efficient Agentic AI Team — June 2025 - Oct. 2025
Research Intern, Amsterdam, Netherlands, Mentors: Davide Belli, Bence Major

- Designed on-device, tool retrieval methods that model query semantics & execution history
- Experimented with prompting methods to balance accuracy & efficiency

Emergence AI — Feb. 2025 – May 2025
Research Intern, Remote, Mentors: Aditya Vempaty, Ashish Jagmohan

- Proposed in-context learning of API functionality from demonstrations for tool-based tasks
- Investigated self-improvement methods with LLM-generated feedback functionality & parameters

GAMMA Lab - University of Maryland, College Park — Aug. 2022 - Present
Graduate Research Assistant, Mentors: Dinesh Manocha, Amrit Singh Bedi

- Exploring the reliability of reference-free LLM judges for prompt optimization
- Investigating memory-augmented LLM agents for personalized embodied agents
- Researching sample-efficient RL training with sparse rewards

Interpretable Machine Learning Lab - Duke University — Jan. 2019 - Mar. 2022
Undergraduate Research Assistant, Mentor: Cynthia Rudin

- Worked on generating hi-res portraits from low-res images with unsupervised representation learning

- Analyzed criminal history data from Broward County, FL, (1.5e5 records), & from Kentucky (3.2e6)

Rein.ai - Remote Mar. 2020-May 2020

Data Science Intern, Mentor: Mohammed Shameer Iqbal

- Set up automated web extraction of truck accident records from 1975-2018 with Python & SQL
- Cleaned & integrated trucking data into the database for the development of risk models

PUBLICATIONS

- *Dynamic Tool Dependency Retrieval for Lightweight Function Calling*
  Bhrij Patel*, Davide Belli*, Amir Jalalirad, Maximilian Arnold, Aleskandr Ermolov, Bence Major
  Association for Computational Linguistics (ACL), 2026
  [Workshop, *Oral*] New Frontiers in Information Retrieval @ Association for the Advancement of Artificial Intelligence (AAAI), 2026
- *Learning API Functionality from In-Context Demonstrations for Tool-based Agents*
  Bhrij Patel, Ashish Jagmohan†, Aditya Vempaty†
  Empirical Methods in Natural Language Processing (EMNLP), 2025
- *Confidence-Controlled Exploration: Efficient Sparse-Reward Policy Learning for Robot Navigation*
  Bhrij Patel, Kasun Weerakoon, Wesley A. Suttle, Alec Koppel, Brian M. Sadler, Tianyi Zhou, Dinesh Manocha, Amrit Singh Bedi
  IEEE/RSJ International Conference on Intelligent Robots & Systems (IROS), 2025
- *Towards Global Optimality for Practical Average Reward Reinforcement Learning without Mixing Time Oracles*
  Bhrij Patel, Wesley A. Suttle, Alec Koppel, Vaneet Aggarwal, Brian M. Sadler, Dinesh Manocha, Amrit Singh Bedi
  International Conference of Machine Learning (ICML), 2024
- *Beyond Exponentially Fast Mixing in Average-Reward Reinforcement Learning via Multi-Level Monte Carlo Actor-Critic*
  Wesley A. Suttle*, Amrit Singh Bedi*, **Bhrij Patel**, Brian M. Sadler, Alec Koppel, Dinesh Manocha
  International Conference of Machine Learning (ICML), 2023
- *In Pursuit of Interpretable, Fair & Accurate Machine Learning for Criminal Recidivism Prediction*
  Caroline Wang*, Bin Han*, **Bhrij Patel**, Cynthia Rudin
  Journal of Quantitative Criminology (JoQC), 2022

  *Denotes Equal Contribution, †Denotes Equal Advising

PREPRINTS & ON-GOING PROJECTS

- *Code Comprehension then Auditing for Unsupervised LLM Evaluation*
  Bhrij Patel, Souradip Chakraborty, Mengdi Wang, Dinesh Manocha, Amrit Singh Bedi
  arXiv, preprint (2026)
- *Personalized Embodied Navigation for Portable Object Finding*
  Vishnu Sashank Dorbala*, **Bhrij Patel***, Amrit Singh Bedi, Dinesh Manocha
  arXiv, preprint (2026)
- *Reward-Free Graph Exploration for Transferable, Embedding-based Retrieval via Successor Representations*
  Bhrij Patel, Vishnu Sashank Dorbala, Alec Koppel, Dinesh Manocha
  Ongoing Project

  *Denotes Equal Contribution

AWARDS

- **2022 University of Maryland, College Park Dean's Fellowship Award**
- **2021 Duke DataFest**: Judges' Pick Award
- **2021 NC State Datathon**: 3rd Place Team
- **2020 COMAP Mathematical Contest in Modeling**: Meritorious Winner
- **2019 Duke University Datathon**: Runner-Up Team

TEACHING

- Teaching Assistant, CMSC 335: Web Application Development with JavaScript, University of Maryland, College Park (Jan 2024-May 2025)
- Teaching Assistant, CMSC 131: Introduction to Object-Oriented Programming, University of Maryland, College Park (Aug-Dec 2023)
- Teaching Assistant, CS 671: Graduate Machine Learning, Duke University (Aug-Dec 2021)
- Teaching Assistant, CS 300: Undergraduate Data Science, Duke University (Jan-May 2021)
- Teaching Assistant, CS 371: Undergraduate Machine Learning, Duke University (Aug-Dec 2020)
- Math Help Room Tutor, Linear Algebra, Duke University (Aug 2019-May 2020)